DAY 12 · THEORY HANDOUT

Transformer Inputs: Tokens and Embeddings

Tokens, embeddings, and positions: the input pieces we need before reading a Transformer diagram.

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1. Tokenization: split the text

A **token** is a text piece. **Tokenization** breaks text into the pieces a model uses.

A cat is playing.



Whole words, word parts, and punctuation can all be tokens.

This split is an illustration. A real tokenizer may split it differently.

A token may be a word, part of a word, or a symbol. A word part is called a subword. Spaces can also be included in tokens.

Word parts help the tokenizer handle many words with a limited list of pieces. Different tokenizers can produce different splits of the same text.

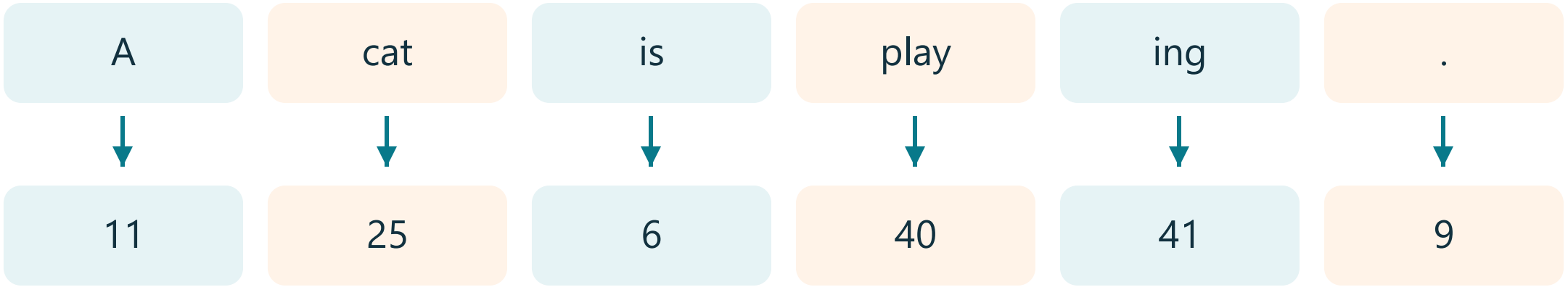
> One token does not always equal one word. The tokenizer prepares pieces; it does not work out their meaning.

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2. Vocabulary and token IDs

A vocabulary is the model tokenizer's token list. Each token has a numeric ID.

Vocabulary = the list of token pieces and their IDs



The tokenizer looks up each piece and returns its ID. The model uses these IDs to find the matching input numbers.

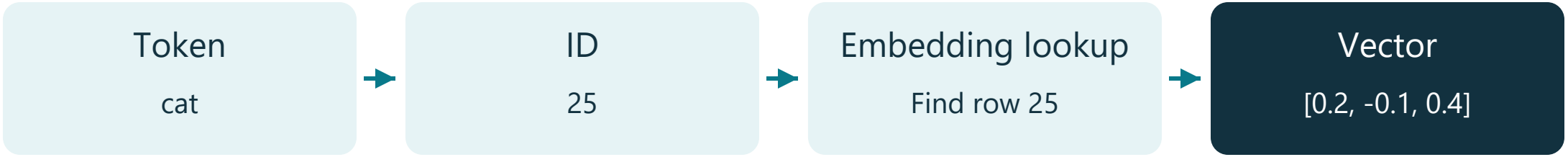
The IDs above are made up. ID 25 means "cat" only in this example. A bigger ID does not mean a smarter, larger, or more positive word.

> Use the tokenizer that belongs to the model. Another tokenizer may use different pieces and IDs.

3. Embedding: a learned row of numbers

An embedding maps a token ID to a vector, a row of numbers the model can work with.

Embedding table = one learned vector for each token ID



Three values are shown only to make the idea visible.

An embedding table stores these vectors. Their values are parameters: numbers changed during training to help the model do its task.

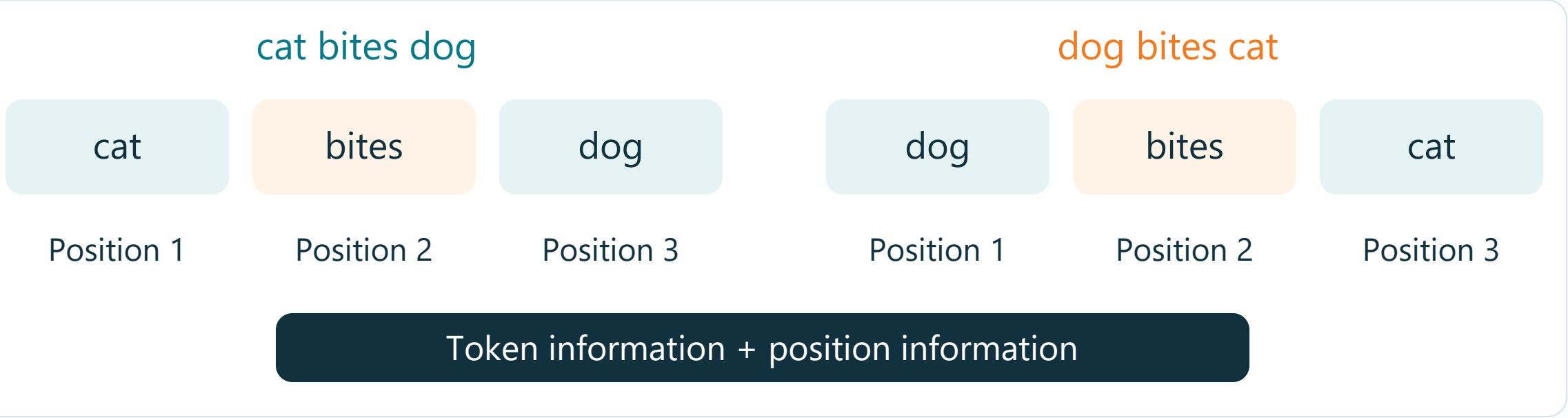
The learned numbers can capture useful patterns of word use. They are not a dictionary definition. Real models usually use far more than three values.

> The vector values above are illustrations. The model learns its own values; we do not choose a meaning number for each word.

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4. Positions keep the word order

Position information tells the model where a token is in the sequence.



The two sentences have the same words but different meaning. Token embeddings alone do not tell us which token comes first.

Transformers use position information as well. In GPT-2, learned position vectors are added to token vectors. Other models may use other methods.

> The goal is the same: the model needs both the token and its place in the text.

5. Context changes the useful information

Context is the surrounding text that helps explain a token.

Same token

"I sat by the river bank."

bank: a place by water.

Different context

"I put money in the bank."

bank: a place for money.

Same input token embedding; later representations can differ.

For the same token ID, the starting embedding is the same. The model then combines information from the available surrounding tokens.

The updated vector is a contextual representation: numbers that now reflect the token and its context. "River" and "money" give different clues.

> The model learns numeric patterns. A useful representation is not proof of human understanding.

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6. What does a Transformer add?

A Transformer is a neural network design that uses attention to work with context.

RNN / LSTM

Carry states step by step.
Earlier information travels
through the sequence.

Transformer

Uses attention to connect
available token information.
Repeats processing blocks.

Attention is a learned way to mix useful information from other available tokens. A block is a set of processing steps repeated in the model.

Within a layer, many input positions can be processed at the same time. This helps training. A text generator still adds its new tokens one at a time.

> Transformer is a design, not a chatbot name. The task depends on the model and how it was trained.

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